

Machine Learning Notes

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Machine Learning (ML)

ML is a computer program which is said to learn from experience E with respect to some class of tasks T and performance measure P .

- Tasks (T): Regression, Decision Tree, etc.
- Experience (E): Training data, Testing data, Validation data.
- Performance Measure (P): F1 Score, Accuracy, Confusion Matrix.

Algorithm vs Machine Learning

Traditional Algorithm: Data + Program $\rightarrow$ Computer $\rightarrow$ Output

Machine Learning: Data + Output (Experience E) $\rightarrow$ Computer $\rightarrow$ Program (Model/Task T)

Additional Notes

ML: Arthur Samuel, 1959.

Basic Components of Learning Process

- Data Storage
- Abstraction
- Generalization
- Evaluation